# ROS topics, state and simulator examples

Developer guide V0.1 | Source software v1.1.0 | 2026-10-11

Shared ROS 2 software interface; B-only services and feedback require the matching supported interface.

Status: Software reference

## From a ROS message to the chassis

Your application publishes a requested velocity. The driver converts that request to the selected chassis protocol and reports state back to ROS. The build and protocol parameters determine which hardware model and message layout it uses.

| ROS interface | Purpose | Scope |
|---|---|---|
| cmd_vel | geometry_msgs/Twist: forward, lateral and yaw-rate request | linear.y is relevant to the A-4WS model |
| ~/drive_command | Explicit motion model and steering request | Supported model depends on build |
| odom and TF | Wheel-based motion estimate and transforms | Not an independent localisation measurement |
| ~/state and ~/safety | Control state, limits and protection feedback | Fields depend on protocol |
| battery_state and /diagnostics | Battery feedback and software diagnostics | Signal scaling must match firmware |
| imu/data_raw | LCI raw acceleration and angular-rate feedback | B draft LCI; no orientation output |

The ~ prefix means the node-private namespace. With the default launch node name, ~/state is /luchs_driver/state. A namespace changes the full name; inspect the actual graph instead of assuming a fixed path.

## Inspect a running simulator

```
ros2 node list
ros2 topic list
ros2 topic echo --once /luchs_driver/state
ros2 topic echo --once /diagnostics
ros2 param dump /luchs_driver
```

Use the quick-start guide to launch dry_run:=true first. State and diagnostics should update. If the node is absent, check the terminal log, workspace environment and build selection.

## A software-only velocity example

Scope note. Run the next example only while the driver was launched with dry_run:=true and is disconnected from physical CAN. It is a simulator illustration, not a vehicle commissioning procedure.

```
ros2 topic pub --rate 10 /cmd_vel geometry_msgs/msg/Twist \
  '{linear: {x: 0.1, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}'
```

The request is 0.1 m/s forward with zero yaw rate. Inspect /odom and /luchs_driver/state in another terminal. Stop the publisher with Ctrl-C; the source driver command watchdog default is 0.5 s. Do not treat that software timeout as a replacement for a physical emergency stop.

## Parameters and B-only services

| Setting | Use |
| --- | --- |
| build / protocol.type | Select the delivered model and communication interface |
| limits.* | Software command limits; do not relabel them as vehicle specifications |
| control.cmd_timeout / feedback_timeout | Source defaults are 0.5 s; retain conservative defaults during setup |
| odom.publish_tf | Disable when a separate estimator owns odom -> base_footprint |

The source lists reset_odometry and set_motion_model services, plus B estop and park requests. B payload-power and speed-limit services are LCI-dependent. Their presence in the API or simulator does not establish that the delivered chassis supports them.

Keep A wheel-order/scaling parameters and B legacy interface settings with the confirmed unit configuration. The B interface guide explains the separate draft and VCU scopes.

## Source and edition

Prepared and reorganised by RJ Tech from the already-public Luchs ROS 2 v1.1.0 package by TerraAgent Robotics. Source licence: Apache-2.0. Source text and examples were shortened, explained and scoped for this topic. Original archives and protocol files remain unchanged.

Source file: README.md

Source file: luchs_driver/README.md

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